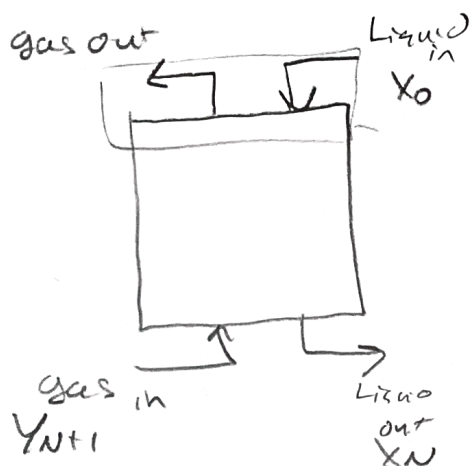


1) What is the stage for x_{out} & y_{out} ?
Start from TOP

Rayan
Baig
EXAM #3
CHE313



entering Liquid
0.0012 mole NB

Desired to remove 91% of NB FROM Liquid

$$L = 100 \frac{\text{kmol}}{\text{min}}$$

Pure vapor
entering Liquid $y_{in} = 0$

$$\frac{L}{V} = 20.0$$

Equilibrium

$$y = 28x$$

$$m = 28$$

$$X_{in} = 0.00012 \text{ NB}$$

$$X_{out} = ?$$

$$Y_{in} = 0$$

$$Y_{out} = ?$$

$$V = 1/20 \frac{\text{kmol}}{\text{min}}$$

A) Desired remove
91% NB FROM Liq
∴

$$X_{out} = X_{in} (1 - 0.91)$$

$$X_{out} = 1.08 \times 10^{-5}$$

Overall mass
Species
bal

$$V Y_{in} + L X_{in} = V Y_{out} + L X_{out}$$

$$Y_{out} = \frac{L}{V} (X_{in} - X_{out})$$

$$Y_{out} = 20 (0.00012 - X_{out})$$

$$Y_{out} = 0.002184$$

B) Number of stages

$$N = \ln \left[\frac{(Y_{N+1} - Y_{N+1}^*)}{(Y_1 - Y_1^*)} \right]$$

$$\ln \left(\frac{L}{mV} \right)$$

$$N = 4.036$$

$$N = 4.036$$

∴

$N = 5$ real
Stages

$$Y_{N+1} = 0 \text{ (Pure liq)}$$

$$Y_1 = Y_{out} = 0.002184$$

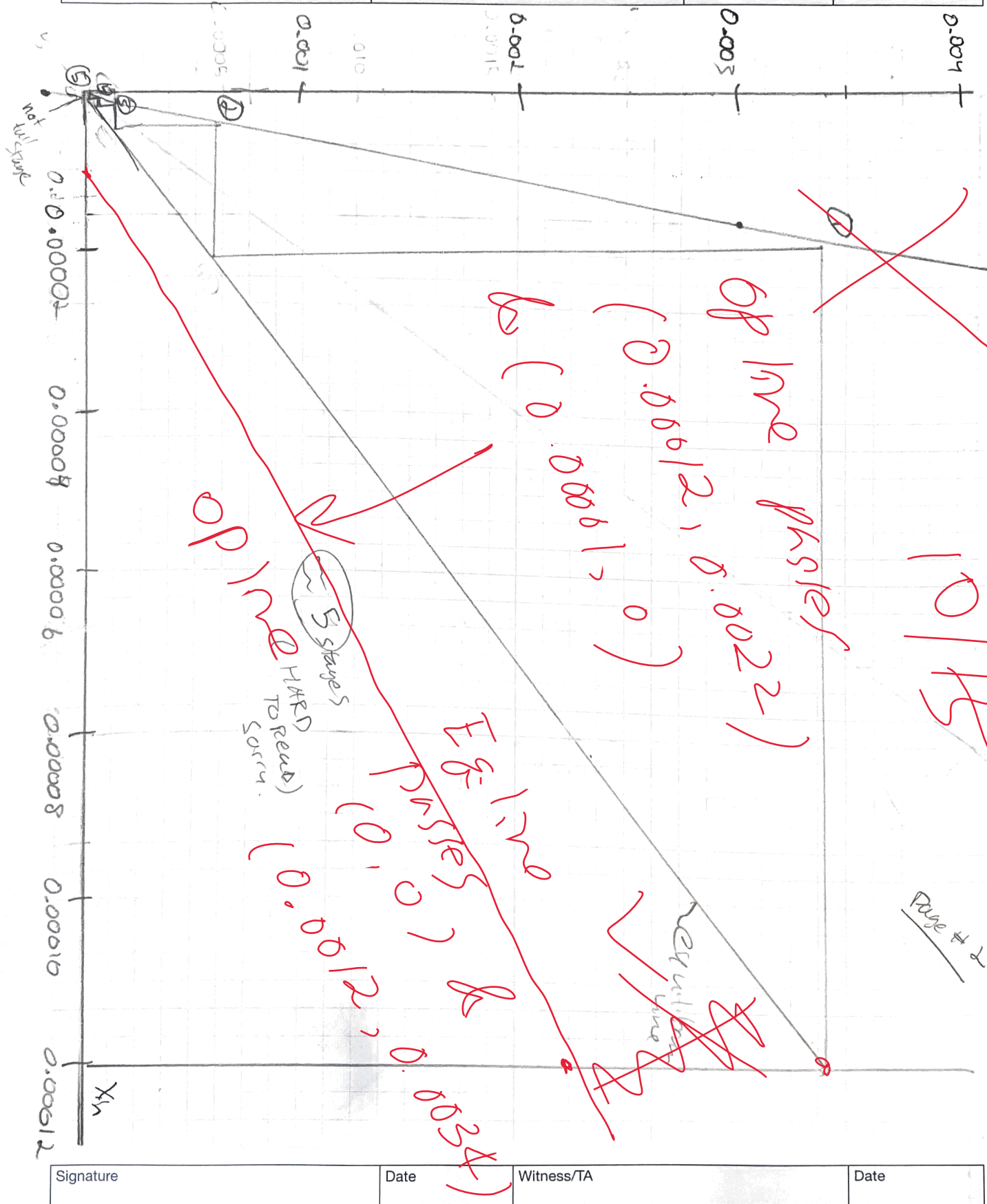
$$Y_{N+1}^* = m X_N = 28 (1.08 \times 10^{-5})$$

$$m = 28$$

$$\frac{L}{V} = 20$$

$$Y_1^* = m X_0 = 28 (0.00012)$$

Exp. No.	Experiment/Subject	Date
Name	Lab Partner	Locker/ Desk No.
		Course & Section No.



operating line

$$V y_{in} + L x = V y + L x_{out}$$

$$L x = V y + L x_{out}$$

$$y = 25x$$

$$x_{in} =$$

$$V y = L (x) - L x_{out}$$

$$y = \frac{L}{V} x - \frac{L}{V} x_{out}$$

$$y = 20x - 20(1.08 \times 10^{-5})$$

$$y = 20x - 0.0000216$$

$$\therefore \text{intersects } y = 1.08 \times 10^{-5} \text{ at } x = 0.002196$$

$$y = 0.003 \text{ @ } x = 1.5 \times 10^{-4}$$

$$V y_{out} + L x = V y + L x_{in}$$

$$y = \left(\begin{matrix} 1.08 \times 10^{-5} \\ 1.2 \times 10^{-4} \end{matrix} \right), \left(\begin{matrix} 0 \\ 0.002196 \end{matrix} \right) \text{ hits the points}$$

$$\text{Slope} = 20$$

$$y_{int} = -0.000216$$

$$y = 20x - 0.000216$$

OP Line

little
over
4 stages = 5
on graph

Problem #2

went 95% of N_2 by air

gas 90.0 mol% N_2
inlet 10.0 mol% N_2

inlet water is pure $\therefore x_{in} = 0$

$$G = 1.0 \text{ mol/h}$$

$$\frac{S}{G} = 45$$

$$y = 28x$$

$$x_0 = 0$$

$$y_2 = 0.90 \quad y_1 = 0.90 \quad x_1 = \frac{0.90}{1 - 0.90} = 0.111$$

$$(x_0, y_1) = (0, 0.111)$$

Mass Bal

$$\cancel{G_1 X + S} \\ \cancel{G_2 Y + X G}$$

want 45% of NB by water
only 0.05 of Y, left

$$Y_{N+1} = Y_1 \times 0.05$$

$$Y_{N+1} = 0.005556$$

$$\cancel{S X + G_2 Y} = \cancel{S X + G_1 Y}$$

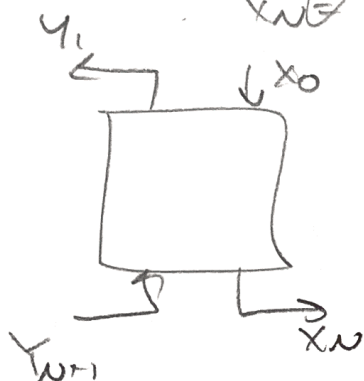
in out

$$\cancel{X_N} = \frac{0.45 \text{ NB}}{0.05 \text{ water}}$$

$$G_2 Y_{N+1} =$$

$$G_2 Y_1 = \cancel{S X_N} + G_1 Y_{N+1}$$

$$X_N = \frac{G_1}{S} (Y_1 - Y_{N+1})$$



$$\cancel{X_0 = 0}$$

$$\cancel{Y_{N+1} = \frac{0.1}{1-0.1} = 0.111}$$

$$\cancel{Y_1 = Y_{N+1} (1-0.95) = 0.005556}$$

In = out

$$Y_{N+1} G + \cancel{S X_0} = S X_N + G_1 Y_1$$

$$X_N = \frac{G_1}{S} (Y_{N+1} - Y_1)$$

$$\cancel{X_N = 0.002346}$$

$$(X_0, Y_1) \rightarrow (0, 0.005556)$$

$$\# (0.002346, 0.111)$$

$$(X_N, Y_{N+1})$$

OP line

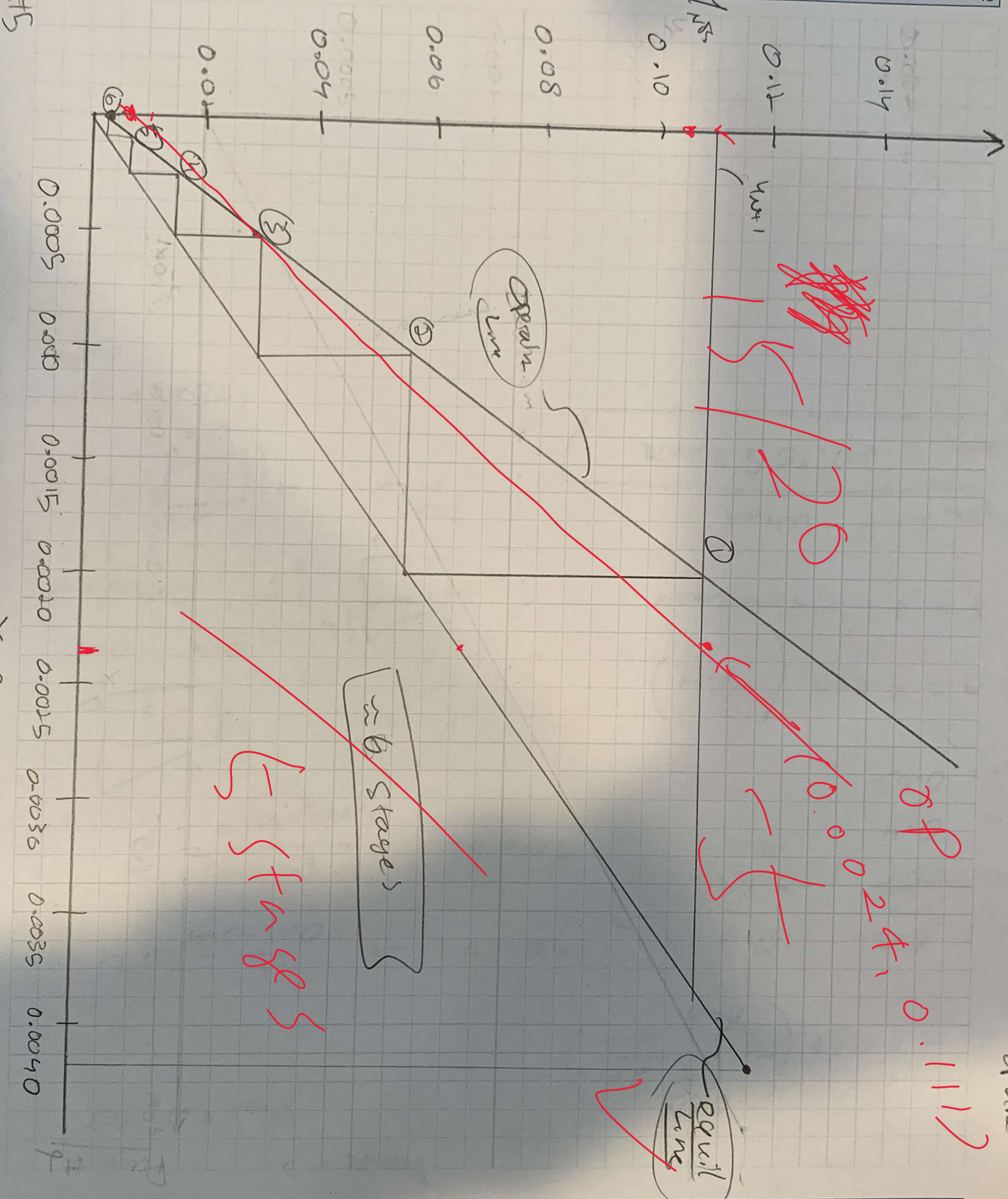
$$y = 45x + [0.005556 - \cancel{45x_0}]$$

20/20

Note: Insert Divider Under Copy Sheet Before Writing

Signature	Date	Witness/TA	Date
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Page #15



Exp. No.	Experiment/Subject	Name
	Lab Partner	
Date	Locker/Desk No.	Course & Section No.

Problem #3

$$W_{fuel} = F_{exp} (-Area)$$

$$\uparrow$$

$$2.0$$

$$2.0 = F_{exp} \left(- \int_{0.40}^{0.80} \frac{2x}{4-x} \right)$$

80% wt A
 & 20% wt B
 Feed

Proton =>

$$F_{fuel} = 0.40$$

~~$x = 0.2$~~ , 0.8
 ~~$y = 0.2$~~ , 0.975
 #2

$$x = 0.4$$

$$\#1 \quad y = 0.65$$

$$2.0 = F_{exp} (-2.4)$$

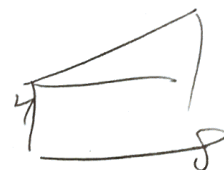
$$F = 22.0464 \text{ kmole}$$

$$P_{TOTAL} = F - W_{fuel}$$

$$P_{TOTAL} = 20.05 \text{ kmole}$$

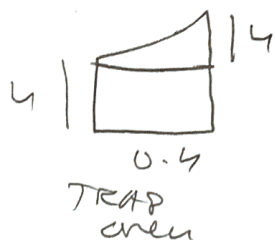
Term #1 $\rightarrow \frac{1}{0.65-0.4} = 4$

Term #2 $\frac{1}{0.975-0.8} = 8$



$$0.4 \quad 0.8$$

$$0.4x(4+8)$$



TRAP
area



$$4(0.4) + \frac{1}{2}(0.4)(4+8)$$

20/20